

Notes: Acemoglu, Aghion, Lelarge, Van Reenen, and Zilibotti (QJE, 2007)

Technology, Information, and the Decentralization of the Firm

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1 Big picture

- **Question.** The paper asks when firms should decentralize authority to lower-level managers rather than keep decisions centralized at headquarters.
- **Core mechanism.** Decentralization gives decision rights to a manager who has superior local information, but it also creates a loss-of-control problem because the manager's objective may not be perfectly aligned with the principal's objective.
- **Technology and information.** The paper links organizational structure to the diffusion of new technologies. When a technology is new, public information about how to implement it is limited, so the manager's private information is valuable. When many firms have already implemented the technology, public information improves and centralization becomes more attractive.
- **Main theoretical predictions.** Firms closer to the technological frontier, firms operating in more heterogeneous environments, and younger firms should be more decentralized.
- **Main empirical setting.** The paper uses three firm/establishment data sets from France and Britain in the 1990s. The main French sample is the COI survey matched to administrative data on the population of French firms.
- **Main decentralization measure.** The baseline measure is whether a firm is organized into profit centers. Profit-center managers have broader authority over revenues, costs, assets, hiring, inventory, and promotion decisions than cost-center or revenue-center managers.
- **Main empirical result.** Firms in more heterogeneous industries, firms closer to their industry productivity frontier, and younger firms are more likely to be decentralized into profit centers.
- **Technology-intensity result.** These relationships are stronger in high-IT or high-tech sectors, consistent with the model's emphasis on the implementation of new technologies.
- **Robustness.** The results survive alternative heterogeneity measures, alternative decentralization proxies, logit/OLS/probit specifications, richer controls, group-firm subsamples, and a British WERS validation sample.
- **Economic takeaway.** Decentralization is not only a matter of firm size or fashion. It is an organizational response to informational uncertainty: when the center has little useful public information, local knowledge becomes valuable enough to justify delegation.

2 Incremental contribution of the paper

- **A new link between technology diffusion and organization.** Earlier work documents that firms have become flatter and more decentralized, but this paper provides a specific theory: decentralization is more valuable when firms face new technologies whose implementation is not yet well understood publicly.
- **Information-based theory of decentralization.** The model puts a clean tradeoff at the center: the manager knows more about the correct action, but delegation gives him scope to act in his own interest.

- **Distance-to-frontier prediction.** The paper generates the novel prediction that firms near the technological frontier should decentralize more because there is less public history from which headquarters can learn.
- **Heterogeneity prediction.** It also predicts that firms in more heterogeneous environments should decentralize more because other firms' experiences are less informative about what will work for the focal firm.
- **Age prediction.** Young firms have accumulated less internal experience, so the principal has less firm-specific knowledge and is more likely to delegate authority.
- **Empirical innovation.** The paper tests these predictions using micro data on actual organizational choices, not merely aggregate trends or case studies.
- **Multiple data-set validation.** The evidence comes from two French data sets and one British data set, with different decentralization measures. This gives the paper more external validity than a single survey would provide.
- **High-tech cross-check.** The paper's strongest results are in high-tech industries, where the theory should be most relevant. This strengthens the interpretation that technology and learning are central mechanisms.
- **Contribution to organizational economics.** The paper gives empirical content to the theoretical literature on delegation, authority, and loss of control, especially the line of work following Aghion and Tirole.

3 Data and measurement

3.1 Main French firm data: COI matched to administrative sources

- The main data set is *Changements Organisationnels et Informatisation* (COI), a survey of French manufacturing firms in 1997.
- The authors match COI firms to FUTE, which covers the population of French firms above employment thresholds and supplies accounting data such as value added, labor productivity, capital, size, market share, and industry information.
- The baseline COI sample contains 3,570 French manufacturing firms after matching.
- The COI is used mainly to measure whether a firm is decentralized into profit centers.

Interpretation: The COI-FUTE match is valuable because it combines organization variables from a survey with productivity and industry variables from administrative data.

3.2 Alternative French data: Enquete Reponse

- The *Enquete Reponse* (ER) is a French establishment-level survey from 1998 covering both manufacturing and nonmanufacturing sectors.
- The paper uses ER for two alternative decentralization measures: managerial autonomy over investment decisions and layering.

- Investment autonomy is measured by whether the plant manager has full or important authority to make investment decisions without headquarters.
- Delaying is measured by whether the establishment reduced the number of hierarchical layers between 1996 and 1998.

Interpretation: ER lets the paper test whether the main findings are specific to profit centers or also appear in direct autonomy and hierarchy measures.

3.3 British validation data: WERS

- The British data come from the 1998 Workplace Employee Relations Survey (WERS).
- WERS contains a measure of plant-manager autonomy over employment decisions: whether the establishment manager can recruit staff without consulting head office.
- Because of confidentiality restrictions, the British analysis has fewer firm-level controls than the French analysis and is restricted to manufacturing establishments.

Interpretation: WERS provides an out-of-country validation. The sample is smaller, but the patterns are broadly consistent with the French evidence.

3.4 Decentralization measures

- **Profit centers.** The baseline COI measure equals one if the firm is organized primarily into profit centers.
- **Investment autonomy.** In ER, decentralization equals one if the plant manager has important or full autonomy over investment decisions.
- **Delaying.** In ER, decentralization is proxied by a reduction in the number of managerial layers.
- **Employment autonomy.** In WERS, decentralization is proxied by whether the establishment manager can make staff-recruitment decisions without headquarters.

Interpretation: Profit-center status is the baseline measure because it captures broad delegation of authority. The alternative measures are noisier but help show that the result is not driven by one survey question.

3.5 Technology, heterogeneity, and age variables

- **Proximity to frontier.** The main measure is the firm's log labor productivity relative to the 99th percentile of labor productivity in its four-digit industry:

$$PF_{it} = \ln y_{it} - \ln y_{it}^F.$$

A higher value means the firm is closer to the frontier.

- **Industry-level heterogeneity in productivity growth.** The benchmark heterogeneity measure is the difference between the 90th and 10th percentiles of firm productivity growth within a four-digit industry:

$$H_l^G = (\Delta \ln y_{il})^{90} - (\Delta \ln y_{il})^{10}.$$

- **Alternative heterogeneity measures.** The paper also uses the 95th-5th spread, the standard deviation of productivity growth, a trimmed standard deviation, dispersion in productivity levels, and firm-level product-space distance.
- **Firm age.** Age is represented by dummies for firms younger than 5 years, 5 to 9 years, 10 to 19 years, and the omitted category of at least 20 years.

Interpretation: The empirical variables map tightly into the theory: proximity to frontier proxies for limited public history; heterogeneity proxies for difficulty learning from others; youth proxies for limited own-firm experience.

4 Models and empirical specifications

4.1 Technology ladder and implementation problem

The model assumes a ladder of technologies $k = 1, 2, \dots$. When firm i gets access to technology k , it must choose one of two implementation actions, L or R . One action is correct for that firm and leads to a successful implementation; the other fails.

The correct action for firm i and technology k is

$$x_{i,k}^* = \begin{cases} x_k^*, & \text{with probability } 1 - \varepsilon, \\ \sim x_k^*, & \text{with probability } \varepsilon, \end{cases}$$

where x_k^* is the reference action for the technology. The parameter ε measures heterogeneity across firms.

- If $\varepsilon = 0$, all firms need the same action.
- If ε is close to $1/2$, other firms' experiences are much less informative for the current firm.
- Successful implementation raises productivity by a factor $\gamma > 1$.

Interpretation: Technology adoption is not just buying a machine. It requires choosing the right way to implement the technology in the firm's specific environment.

4.2 Centralization versus delegation

The firm can choose between two organizational forms:

- **Centralization:** the principal chooses the action using public information from previous implementations by other firms.
- **Delegation:** the manager chooses the action using superior private information about what will work in the firm.

The manager observes the correct action but may prefer another action. His preferred action is

$$z_{i,k}^* = \begin{cases} x_{i,k}^*, & \text{with probability } \delta, \\ \sim x_{i,k}^*, & \text{with probability } 1 - \delta. \end{cases}$$

The parameter δ measures congruence between the principal and manager. The paper assumes $\delta \in (1/2, 1 - \varepsilon)$.

Interpretation: Delegation is valuable because the manager knows more, but costly because the manager may use his information to choose an action that benefits him rather than the principal.

4.3 Learning from public history

Let $h_k^i = \{n_k^i, \tilde{n}_k^i\}$ summarize the public history for technology k before firm i acts:

- n_k^i is the number of previous firms that have attempted technology k .
- $\tilde{n}_k^i$ is the number of previous attempts for which L turned out to be profitable.

Under delegation, the probability of a correct action is $\pi(1; h_k^i) = \delta$. Under centralization, the probability is $\pi(0; h_k^i)$ and depends on the public history.

- At the frontier, $n_k^i = 0$, so the principal has no useful public information and $\pi(0; h_k^i) = 1/2$.
- Far behind the frontier, n_k^i is large, so the principal can learn the reference action from others and $\pi(0; h_k^i)$ approaches $1 - \varepsilon$.

Interpretation: Distance to the frontier matters because it changes the amount of information that has already accumulated in the public domain.

4.4 Theoretical predictions

- **Prediction 1: proximity to frontier.** Firms at the technological frontier should delegate because headquarters cannot learn from previous adopters. Firms far behind should centralize because public information is abundant.
- **Prediction 2: heterogeneity.** Holding distance to frontier fixed, greater heterogeneity makes delegation more likely because other firms' outcomes are less informative.
- **Prediction 3: age.** Younger firms should decentralize more because they have less internal experience from which the principal can learn.

Interpretation: The theory is not simply that all advanced firms decentralize. It is that decentralization becomes attractive when the principal's information set is weak relative to the manager's local knowledge.

4.5 Baseline empirical specification

The empirical model is a latent-index model for decentralization:

$$d_{ilt}^* = \alpha H_{ilt-1} + \beta P F_{ilt-1} + \gamma age_{ilt-1} + w'_{ilt-1} \zeta + u_{ilt},$$

where i indexes firms, l industries, and t time. The observed decentralization dummy is

$$d_{ilt} = 1\{d_{ilt}^* > 0\}.$$

The paper estimates the model mainly by maximum-likelihood probit and reports marginal effects.

The authors also estimate an unconstrained version that separates the firm’s own productivity from the frontier productivity:

$$d_{ilt}^* = \alpha H_{ilt-1} + \beta_1 \ln y_{ilt-1} + \beta_2 \ln y_{ilt-1}^F + \gamma age_{ilt-1} + w'_{ilt-1} \zeta + u_{ilt}.$$

Interpretation: The unconstrained version helps address the concern that decentralization might simply raise the firm’s own productivity. If the frontier term also enters with the predicted opposite sign, the evidence better matches the theory.

5 Identification and empirical design

5.1 What the empirical design can identify

- The paper documents robust correlations between decentralization and three theoretically motivated variables: heterogeneity, proximity to frontier, and age.
- The authors explicitly do not claim that the baseline regressions identify fully causal effects.
- The empirical design is strongest as evidence that the comparative statics of the theory appear in actual organizational data.

Interpretation: The right reading is not “moving a firm closer to the frontier mechanically causes decentralization.” Rather, firms facing frontier-like information problems are systematically more likely to decentralize.

5.2 Addressing reverse causality and omitted variables

- Right-hand-side variables are lagged and averaged over 1994-1997, while the decentralization variable is measured in 1997 or 1998, reducing direct simultaneity concerns.
- Regressions include three-digit industry dummies and a large set of firm and industry controls.
- The key controls include firm size, number of plants, computer use, skill share, capital intensity, foreign ownership, worker age, product market competition, market share, specialization, Herfindahl index, industry capital, and industry IT intensity.
- Standard errors are robust and clustered at the four-digit industry level.

Interpretation: The paper tries to rule out simple confounds such as size, IT use, capital intensity, foreign ownership, and industry composition.

5.3 Robustness strategy

- **Alternative heterogeneity measures:** the main results are compared across growth dispersion, level dispersion, standard deviations, trimmed measures, and firm-level product-space distance.
- **Alternative decentralization measures:** the paper repeats the analysis using investment autonomy, delayering, and British employment autonomy.
- **Alternative productivity measures:** the authors report similar findings using TFP and alternative definitions of the frontier, such as the 95th and 90th percentiles.
- **High-tech split:** the theory predicts stronger results where new technology implementation is more important; the high-tech estimates are indeed stronger.
- **Instrumental-variable checks:** the authors use British industry counterparts as instruments for French industry heterogeneity and frontier variables. These estimates remain positive and significant.

Interpretation: The empirical strategy is cumulative. No single regression proves causality, but the repeated consistency across measures and countries strengthens the informational-learning interpretation.

5.4 Main limitations

- The decentralization data are mostly cross-sectional, so the paper cannot fully distinguish firm age effects from cohort effects.
- Profit-center organization may partly reflect firm complexity or accounting practices, although many controls address this concern.
- Firm productivity may itself be affected by decentralization, making the proximity-to-frontier regressions potentially endogenous.
- The UK sample is small and lacks the rich firm-level controls available in the French data.
- Some alternative measures, especially age effects with delayering and investment autonomy, are weaker than the baseline results.

Interpretation: The evidence is broad and carefully checked, but the paper remains mainly a theory-guided empirical correlation study rather than a randomized or quasi-experimental design.

6 Tables and figures

6.1 Figures I–III: Basic patterns

Read:

- Figure I groups firms by industry heterogeneity. The share of firms decentralized into profit centers generally rises as heterogeneity increases; it reaches about 41% in the tenth decile.

- Figure II groups firms by proximity to the productivity frontier. Firms closer to the frontier are more likely to be decentralized; the top deciles show noticeably higher decentralization rates than the lower deciles.
- Figure III groups firms by age. Firms younger than five years are much more likely to be decentralized than older firms.
- These figures are descriptive and do not include controls, but they motivate the theory and the regression design.

Economic message: The raw data already show the three patterns predicted by the model: more decentralization under heterogeneity, near the frontier, and among young firms.

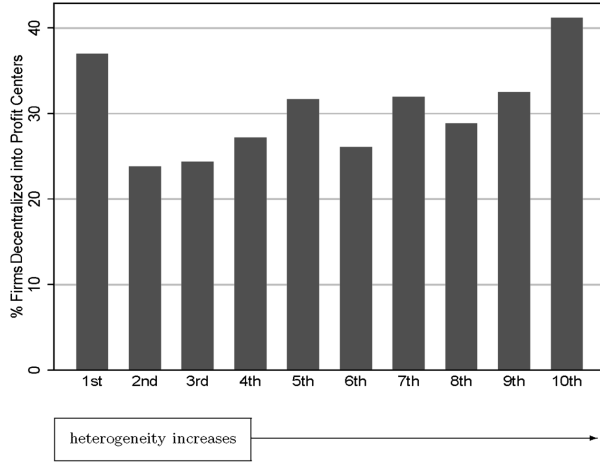


FIGURE I
Heterogeneity and Decentralization

Notes. The X-axis divides all firms into deciles of heterogeneity from the first decile (low heterogeneity) to the tenth decile (high heterogeneity). Heterogeneity is measured by an index of dispersion (the difference of the 90th minus the 10th percentile) of productivity growth between firms in a four-digit industry (see text). The Y-axis indicates the proportion of firms that are decentralized into profit centers in the relevant decile group. The sample is the COI sample (3,570 French firms in 1997).

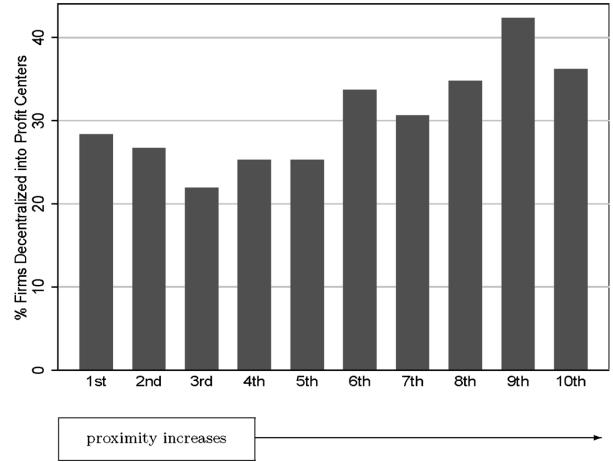


FIGURE II
Proximity to Frontier and Decentralization

Notes. The X-axis divides all firms into deciles of proximity to frontier (in terms of value added per hour) from the first decile (low proximity to frontier) to the tenth decile (high proximity to frontier). The Y-axis indicates the proportion of firms that are decentralized into profit centers in the relevant decile group. The sample is the COI sample (3,570 French firms in 1997).

(a) Figure I: Heterogeneity and decentralization.

(b) Figure II: Proximity to frontier and decentralization.

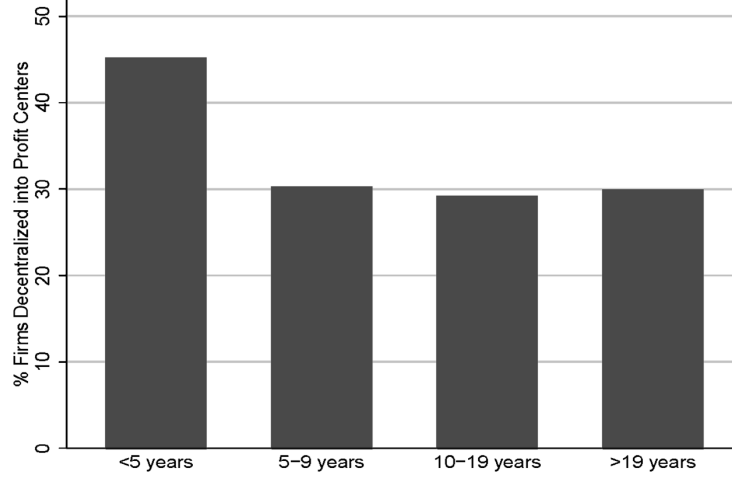


FIGURE III

Firm Age and Decentralization

Notes. Firms are grouped into age bands (dated from the birth of the firm). The Y-axis indicates the proportion of firms that are decentralized into profit centers in the relevant age group. The sample is the COI sample (3,570 French firms in 1997).

Figure III: Firm age and decentralization.

6.2 Table I: Descriptive statistics

Read:

- Table I describes the variables used in the core French analysis.
- The mean of the baseline decentralization measure is 0.304, meaning about 30% of COI firms are organized into profit centers.
- The ER alternative measures show that 48.4% of establishments have decentralized investment decisions and 43.6% experienced layering.
- The benchmark heterogeneity measure, the 90th-10th percentile spread of labor productivity growth, has a mean of 0.275.
- The mean proximity-to-frontier measure is 0.358, and the mean log proximity is -1.125.
- The average firm has about 323 workers, three plants, and is about 22 years old.

Economic message: The table clarifies that the paper studies economically meaningful variation: decentralization is common but far from universal, and there is substantial dispersion in productivity, age, computer use, size, and industry environment.

TABLE I
DESCRIPTIVE STATISTICS (ENQUÊTE COI, 1994–1997)

Variable	Source	Mean	Median	St dev
Decentralization measures				
Organized into profit centers	COI	0.304	0	0.460
Decentralization of investment decisions	ER	0.484	0	0.500
Delaying	ER	0.436	0	0.496
Heterogeneity measures: Industry-level heterogeneity of labor productivity growth (H_t^P)				
90th – 10th percentiles	DADS/FUTE	0.275	0.263	0.087
95th – 5th percentiles	DADS/FUTE	0.443	0.406	0.160
Standard deviation	DADS/FUTE	0.177	0.165	0.066
Standard deviation after trimming	DADS/FUTE	0.088	0.082	0.033
Heterogeneity measures: Industry-level heterogeneity of labor productivity level (H_t^L)				
90th – 10th percentiles	DADS/FUTE	0.897	0.861	0.229
Heterogeneity measure: Firm-level inverse share of close firms in the product space (H_t^F)				
Share of close (IT-weighted) firms, %	FUTE	0.343	0.138	0.049
In (firm-specific heterogeneity)	FUTE	7.111	6.587	2.381
Share of close (unweighted) firms, %	FUTE	0.216	0.096	0.308
Distance to technological frontier				
Firm labor productivity	DADS/FUTE	0.163	0.143	0.089
Sectoral 99th percentile labor productivity	DADS/FUTE	0.508	0.397	0.315
Proximity to frontier	DADS/FUTE	0.358	0.334	0.159
In (proximity to frontier)		-1.125	-1.096	0.457
Other firm-level variables				
Foreign ownership	LIFI	0.173	0	0.379
Firms belonging to a larger group	LIFI	0.547	1	0.498

TABLE I
(CONTINUED)

Variable	Source	Mean	Median	St dev
Number of plants	DADS	3.092	1	8.510
Firm age	SIRENE	21.658	18	12.740
Capital stock/value added	FUTE	1.143	0.907	1.036
Number of workers	FUTE	323.463	88.375	677.080
Percentage of workers using computers	COI	59.669	71.846	26.300
Percentage of skilled workers	DADS	72.996	77.371	20.202
Age of workers	DADS	38.870	39.010	3.403
Lerner index	FUTE	0.075	0.068	0.077
Market share (%)	FUTE	1.732	0.404	4.171
Herfindahl index	FUTE	0.049	0.031	0.057
Specialization	FUTE	0.831	0.931	0.203
Other industry-level variables				
Capital stock per worker	BRN	404.987	289.242	369.064
IT investment (per worker)	EAE/FUTE	0.849	0.600	0.725

Note: These descriptive statistics are based on the COI sample (3,570 observations), except for “Decentralization of investment decisions” and “Delaying” (ER sample: 1,858 observations). The COI data set is a firm-level survey providing information on organization and other firm characteristics in 1997; it covers manufacturing sector only. The ER data set is an establishment survey containing information about organizational change between 1996 and 1998; it covers both manufacturing and nonmanufacturing sectors. The FUTE files contain the firms’ balance sheets and further accounting information; it refers to the entire population of French firms having more than 30 employees in manufacturing industries or more than 10 employees in other industries. The DADS files consist of yearly mandatory employer reports of each worker’s hours (and gross earnings subject to payroll taxes); they cover the entire population of French firms. The LIFI files describe the structure of ownership of large French firms; it also includes information about their main interests in other companies. SIRENE is the register of all French firms. Units of currency are thousands of French francs at the 1995 prices. See text for variable definitions.

(a) Table I, part 1: decentralization, heterogeneity, and frontier variables. (b) Table I, part 2: firm and industry controls.

6.3 Table II: Determinants of decentralization in the COI sample

Read:

- Table II is the core regression table using profit-center decentralization as the dependent variable.
- Heterogeneity has a positive marginal effect. In column 4, the marginal effect is 0.251 with standard error 0.115.
- Proximity to frontier is positive and significant. In column 4, the marginal effect is 0.164 with standard error 0.028.
- Young firms are more decentralized. In column 4, firms younger than five years are 17.4 percentage points more likely to be decentralized than firms at least 20 years old.
- More competitive firms are more decentralized: the Lerner index is negative and significant.
- High-tech sectors show much stronger patterns. Heterogeneity has a large positive effect in high-tech industries but not in low-tech industries.

Economic message: The main regression evidence matches the theory. Decentralization is most likely where headquarters has less reliable public information: high heterogeneity, frontier proximity, and firm youth.

Industries:	Dependent variable: Firm decentralized into profit centers					
	Full sample			High-tech	Low-tech	
	(1)	(2)	(3)	(4)	(5)	(6)
Heterogeneity (H^D)	0.211 (0.107)	0.252 (0.102)	0.296 (0.127)	0.251 (0.115)	0.679 (0.189)	-0.062 (0.135)
Frontier, 99th percentile ($\ln y_{99}$)	-0.101 (0.039)	—	0.225 (0.045)	—	—	—
Labor productivity, firm-level ($\ln y_i$)	0.182 (0.028)	—	0.141 (0.033)	—	—	—
Proximity to frontier (constrained term $\ln y_i - \ln y_{99}$)	—	0.167 (0.024)	—	0.164 (0.028)	0.224 (0.040)	0.103 (0.044)
Firm age < 5 years	0.151 (0.040)	0.151 (0.040)	0.172 (0.041)	0.174 (0.041)	0.215 (0.059)	0.122 (0.055)
5 ≤ firm age < 10 years	0.012 (0.021)	0.012 (0.021)	0.066 (0.022)	0.060 (0.022)	0.069 (0.032)	0.049 (0.029)
10 ≤ firm age < 20 years	-0.007 (0.019)	-0.007 (0.019)	0.039 (0.019)	0.040 (0.019)	-0.008 (0.027)	0.083 (0.026)
Lerner index	—	—	-0.660 (0.144)	-0.733 (0.136)	-0.947 (0.168)	-0.547 (0.218)
In (number of plants)	—	—	0.041 (0.015)	0.041 (0.015)	0.027 (0.021)	0.052 (0.017)
In (firm size)	—	—	0.110 (0.009)	0.110 (0.009)	0.134 (0.015)	0.098 (0.014)
Percentage of workers using computers	—	—	0.220 (0.034)	0.217 (0.034)	0.238 (0.071)	0.189 (0.034)

Industries:	Dependent variable: Firm decentralized into profit centers					
	Full sample			High-tech	Low-tech	
	(1)	(2)	(3)	(4)	(5)	(6)
Percentage of skilled workers	—	—	0.169 (0.049)	0.153 (0.048)	0.206 (0.078)	0.090 (0.039)
In (firm capital/value added)	—	—	0.008 (0.012)	0.009 (0.012)	-0.003 (0.017)	0.022 (0.016)
Foreign ownership	—	—	0.047 (0.025)	0.045 (0.025)	0.064 (0.038)	0.015 (0.032)
Average age of workers in the firm (/10)	—	—	-0.057 (0.025)	-0.060 (0.025)	-0.155 (0.046)	0.008 (0.027)
Firm market share	—	—	-0.577 (0.260)	-0.574 (0.257)	-0.821 (0.382)	-0.520 (0.419)
In (firm specialization)	—	—	-0.071 (0.030)	-0.070 (0.030)	-0.119 (0.047)	-0.027 (0.039)
In (Herfindahl index)	—	—	-0.015 (0.017)	-0.011 (0.017)	0.031 (0.027)	-0.024 (0.022)
In (sector capital stock per worker)	—	—	-0.064 (0.040)	-0.074 (0.040)	-0.115 (0.035)	-0.033 (0.041)
In (sector IT investment per worker)	—	—	0.116 (0.019)	0.102 (0.019)	0.059 (0.044)	0.111 (0.029)
Industry dummies	yes (73)	yes (73)	yes (73)	yes (73)	yes (52)	yes (42)
Mean of dependent variable	0.304	0.304	0.304	0.304	0.373	0.232
Observations	3,570	3,570	3,570	3,570	1,767	1,803

Note. All coefficients are marginal effects from probit maximum likelihood estimation. Robust standard errors corrected for arbitrary variance-covariance matrix at the four-digit industry level in parentheses. Industry variables are defined at the four-digit level (except industry dummies which are at the three-digit level). All right-hand-side variables are lagged and averaged over three years (1984–1997). Labor productivity is the log of value added per hour; frontier is defined as the 99th percentile of the productivity distribution in the firm's four-digit industry. Heterogeneity is defined as the dispersion of productivity growth rate within a four-digit industry (the 99th percentile minus the 10th percentile). The unitized category for firm age is "firm age greater than twenty years." High-tech subsample includes all firms in industries with greater than median IT investment per worker. Low-tech subsample includes all firms in industries with less than median IT investment per worker. See text for variable definitions.

(a) Table II, part 1: main variables and first controls.

(b) Table II, part 2: additional controls and sample details.

6.4 Table III: Alternative measures of heterogeneity

Read:

- Table III replaces the benchmark heterogeneity measure with several alternatives.
- The 95th-5th percentile spread, the standard deviation of productivity growth, and the trimmed standard deviation all produce positive heterogeneity effects.
- The level-based heterogeneity measure is weaker in the full sample but strong and significant in the high-tech sample.
- The firm-specific IT-weighted product-space heterogeneity index is positive and significant in the full sample, and its effect is stronger in high-tech industries.
- Proximity to frontier and young-firm effects remain broadly stable across these specifications.

Economic message: The heterogeneity result is not an artifact of one exact definition. Measures that capture either industry dispersion or firm-specific lack of close neighbors point in the same direction, especially in high-tech sectors.

TABLE III
DETERMINANTS OF DECENTRALIZATION: ALTERNATIVE MEASURES OF HETEROGENEITY (ENQUÊTE COI)

Measure of heterogeneity	Dependent variable (mean = 0.304): Firm decentralized into profit centers										
	Heterogeneity: dispersion of productivity <i>growth</i> in four-digit industry (H_1^G)				Heterogeneity: dispersion of productivity <i>levels</i> in four-digit industry (H_1^L)			Heterogeneity: ln of the inverse share of "close" firms in the product space (U_{10} , H_1^P)			
	Std dev after trimming										
	90th – 10th percentiles	95th – 5th percentiles	Standard deviation		90th – 10th percentiles			IT-weighted			Unweighted
Industry	(1)	Full sample (2)	(3)	(4)	Full sample (5)	High-tech (6)	Low-tech (7)	Full sample (8)	High-tech (9)	Low-tech (10)	Full sample (11)
Heterogeneity	0.251 (0.115)	0.142 (0.069)	0.231 (0.132)	0.870 (0.316)	0.090 (0.062)	0.271 (0.092)	–0.064 (0.090)	0.063 (0.031)	0.098 (0.048)	0.019 (0.037)	0.055 (0.035)
Proximity to frontier	0.164 (0.028)	0.168 (0.029)	0.152 (0.028)	0.165 (0.028)	0.153 (0.028)	0.222 (0.041)	0.100 (0.044)	0.159 (0.028)	0.208 (0.039)	0.104 (0.043)	0.146 (0.027)
Firm age < 5 years	0.174 (0.041)	0.173 (0.041)	0.171 (0.041)	0.173 (0.041)	0.171 (0.041)	0.211 (0.059)	0.121 (0.056)	0.177 (0.041)	0.214 (0.060)	0.123 (0.056)	0.173 (0.041)
5 ≤ firm age < 10 years	0.066 (0.022)	0.067 (0.022)	0.067 (0.022)	0.066 (0.022)	0.068 (0.022)	0.067 (0.032)	0.049 (0.029)	0.067 (0.022)	0.068 (0.032)	0.049 (0.029)	0.067 (0.022)
10 ≤ firm age < 20 years	0.040 (0.019)	0.041 (0.019)	0.040 (0.019)	0.040 (0.019)	0.039 (0.019)	–0.009 (0.027)	0.083 (0.028)	0.041 (0.019)	–0.005 (0.027)	0.082 (0.028)	0.040 (0.019)
Lerner index	–0.733 (0.136)	–0.750 (0.137)	–0.708 (0.134)	–0.743 (0.136)	–0.725 (0.132)	–0.952 (0.172)	–0.534 (0.217)	–0.721 (0.136)	–0.895 (0.168)	–0.544 (0.218)	–0.697 (0.133)
Industry dummies	yes (73)	yes (73)	yes (73)	yes (73)	yes (73)	yes (52)	yes (42)	yes (73)	yes (52)	yes (42)	yes (73)
Observations	3,570	3,570	3,570	3,570	3,570	1,767	1,803	3,570	1,767	1,803	3,570

Note. All coefficients are marginal effects from probit maximum likelihood estimation. Robust standard errors corrected for arbitrary variance-covariance matrix at the four-digit industry level in parentheses. In column (1) the measure of heterogeneity is the difference between the productivity *growth rate* for the firm at the 90th percentile minus the 10th percentile; in column (2) it is the 95th percentile minus the 5th percentile; in column (3) it is the standard deviation of productivity growth rates and in column (4) it is the same as column (2) except that we trim the bottom and top 5% of productivity growth distribution in each four-digit industry. In columns (5)–(7) the measure of heterogeneity is the difference

Table III: Alternative measures of heterogeneity.

6.5 Tables IV and V: Alternative decentralization measures and Britain

Read:

- Table IV uses ER measures: investment-decision autonomy and layering.
- For investment autonomy, frontier productivity is negatively related to decentralization, meaning firms closer to the frontier are more decentralized.
- Heterogeneity is especially positive in the high-tech sample for investment autonomy.
- For layering, productivity variables and heterogeneity generally have the predicted signs, although some estimates are less precise.
- Table V uses British WERS data and employment-decision autonomy. Heterogeneity is positive and significant, and the frontier term is negative in the richer specifications.
- The British results are important because they replicate the broad pattern outside France, even with fewer controls and a smaller sample.

Economic message: The main story is not tied only to profit centers. Measures of actual decision autonomy and changes in hierarchy point in a similar direction, though with noisier estimates.

TABLE IV
DETERMINANTS OF DECENTRALIZATION: ALTERNATIVE MEASURES OF DECENTRALIZATION (ENQUÊTE REPONSE)

Dependent variable Specification of proximity to frontier	Decentralization of investment decisions, 1998			Some delaying between 1996 and 1998					
	Level			Level			Growth rate		
	Full sample (1)	High- tech (2)	Low- tech (3)	Full sample (4)	High- tech (5)	Low- tech (6)	Full sample (7)	High- tech (8)	Low- tech (9)
Heterogeneity (H^C)	0.108 (0.165)	0.455 (0.235)	-0.127 (0.415)	0.355 (0.185)	0.398 (0.283)	0.075 (0.301)	0.434 (0.200)	0.396 (0.304)	0.102 (0.284)
Frontier (99th percentile)	-0.110 (0.054)	-0.249 (0.079)	0.021 (0.103)	-0.115 (0.051)	-0.152 (0.080)	0.011 (0.086)	-0.058 (0.048)	-0.078 (0.064)	-0.043 (0.067)
Labor productivity (firms)	0.072 (0.052)	0.064 (0.077)	0.104 (0.075)	0.126 (0.056)	0.118 (0.072)	0.066 (0.081)	-0.056 (0.101)	0.164 (0.157)	-0.137 (0.109)
Firm age < 5 years	-0.036 (0.056)	-0.016 (0.080)	-0.027 (0.092)	0.183 (0.088)	0.099 (0.113)	0.275 (0.129)	0.054 (0.107)	0.102 (0.171)	0.017 (0.114)
5 ≤ firm age < 10 years	-0.007 (0.048)	-0.044 (0.074)	0.022 (0.061)	0.047 (0.056)	-0.007 (0.064)	0.076 (0.084)	0.034 (0.059)	-0.031 (0.068)	0.095 (0.090)
10 ≤ firm age < 20 years	0.004 (0.040)	-0.015 (0.059)	0.026 (0.054)	0.069 (0.039)	0.059 (0.054)	0.068 (0.055)	0.072 (0.040)	0.056 (0.053)	0.099 (0.054)
Lerner index	-0.034 (0.009)	-0.046 (0.024)	-0.029 (0.114)	-0.934 (0.288)	-0.166 (0.416)	-0.981 (0.402)	-0.666 (0.257)	0.048 (0.358)	-0.977 (0.359)
Industry dummies	yes (81)	yes (49)	yes (31)	yes (70)	yes (47)	yes (43)	yes (69)	yes (47)	yes (42)
Mean of dependent variable	0.484	0.475	0.493	0.440	0.553	0.325	0.436	0.553	0.320
Observations	1,258	648	610	1,049	526	523	1,011	505	501

Note. All coefficients are marginal effects from probit maximum likelihood estimation. Robust standard errors corrected for arbitrary variance-covariance matrix at the four-digit industry level in parentheses. All establishments are part of a large (French or foreign) group but are not headquarters. Heterogeneity is defined as the dispersion of productivity growth rates within a four-digit industry (the 90th percentile minus the 10th percentile). Full set of firm- and industry-level controls included as in Table II, column (1); see text for variable definitions.

TABLE V
DETERMINANTS OF DECENTRALIZATION IN BRITAIN (BRITISH WERS98)

Dependent variable (mean = 0.805)	Decentralization of employment decisions					
	(1)	(2)	(3)	(4)	(5)	(6)
Heterogeneity (H^C)	0.273 (0.130)	—	—	—	0.316 (0.129)	—
Heterogeneity (H^C)	—	0.540 (0.325)	—	—	—	0.659 (0.312)
Frontier (99th percentile)	—	—	-0.051 (0.073)	—	-0.204 (0.073)	-0.156 (0.073)
Establishment age < 5 years	—	—	—	-0.076 (0.115)	-0.069 (0.116)	-0.123 (0.114)
5 ≤ est. age < 10 years	—	—	—	0.086 (0.081)	0.055 (0.089)	0.049 (0.089)
10 ≤ est. age < 20 years	—	—	—	-0.127 (0.077)	-0.164 (0.076)	-0.173 (0.075)
Many competitors	—	—	—	—	0.127 (0.082)	0.150 (0.078)
Few competitors	—	—	—	—	0.210 (0.070)	0.228 (0.065)
No competitors	—	—	—	—	ref	ref
Other establishment controls	no	no	no	no	yes	yes
Industry dummies	yes (64)	yes (64)	yes (64)	yes (64)	yes (64)	yes (64)
Observations	236	236	236	236	236	236

Note. All coefficients are marginal effects from linear probability models. Robust standard errors corrected for arbitrary variance-covariance matrix at the four-digit industry level in parentheses. Data are from the 1998 British Workplace Employee Relations Survey (WERS); they include manufacturing establishments only. Dependent variable is a dummy variable indicating whether "Establishment's manager is able to make decisions on which staff to recruit without consulting Head Office." Heterogeneity and frontier are averaged between 1994 and 1997. All regressions include a control for employment size (current, lagged one year, and lagged five years). Other establishment controls include the proportion of young workers (under 20 years old), the proportion of older workers (aged over 50 years old), the proportion of unskilled manual workers, and the proportion of part-time workers. See text for variable definitions.

(a) Table IV: Alternative decentralization measures in France.

(b) Table V: Decentralization in Britain.

7 Short summary

- The paper develops a model where firms decide between centralization and delegation when implementing new technologies.
- Centralization uses public information; delegation uses the manager's superior local information.
- Delegation is costly because the manager may have objectives that differ from the principal's.
- Public information accumulates as other firms implement a technology.
- Therefore, firms at the frontier, firms in heterogeneous environments, and young firms should decentralize more.
- The main French COI evidence supports all three predictions using profit-center organization as the decentralization measure.
- Effects are economically large: doubling heterogeneity increases the predicted probability of decentralization by about 7 percentage points from a 30% base.
- The relationships are stronger in high-tech industries, exactly where technology implementation and learning should matter most.
- Alternative heterogeneity measures and alternative decentralization measures generally support the same interpretation.
- British WERS evidence gives additional cross-country support.
- The paper is best read as theory-guided evidence on robust organizational patterns, not as a fully causal design.

8 Conclusion

8.1 Core conclusion

Acemoglu, Aghion, Lelarge, Van Reenen, and Zilibotti show that decentralization is systematically related to technology, information, and learning. Firms decentralize when local managerial

information is especially valuable relative to what headquarters can learn from public or internal history.

8.2 Economic intuition

The economic intuition is simple. If many firms have already implemented a technology, headquarters can infer the right implementation action from public history, so central control becomes attractive. If the firm is near the frontier, in a heterogeneous environment, or young, public and internal histories are less useful. In those cases, the local manager's information becomes more valuable, and delegation is more likely even though it creates agency costs.

8.3 Incremental contribution revisited

The paper advances the literature in two ways. First, it provides a tractable model linking the internal organization of the firm to technology diffusion and information accumulation. Second, it brings this model to micro data using firm-level and establishment-level measures of decentralization in France and Britain.

8.4 Implications for organizational design

The findings imply that decentralization is not universally optimal. It depends on the informational environment. Firms should be more willing to delegate when they face novel, uncertain, or firm-specific implementation problems. Firms can centralize more when learning from others or from their own past experience gives headquarters reliable information.

8.5 Limitations

The evidence is largely observational. Productivity, frontier position, and decentralization may be jointly determined. The profit-center measure is strong but not perfect, and alternative measures produce somewhat noisier results. The age result is particularly less robust outside the baseline COI specification. The British sample is small and less richly controlled.

8.6 Takeaway

The enduring takeaway is that the structure of the firm reflects the structure of information. Decentralization is more likely when firms operate close to the technological frontier, face heterogeneous environments, or lack accumulated experience. The paper turns the broad idea that technology changes organization into a concrete set of testable predictions about when firms delegate authority.